

Note:

- i. Answer all the questions by choosing any one of the sub questions.
 ii. Each question carries 10 marks

(10 marks)		CO1
Q1	Define three- schema architecture of database system. A hospital stores patient information in a database. The database administrator changes the storage device and indexing method to improve query performance. Doctors and nurses continue using the same applications.	
(a)	a) Which level of architecture changed? b) Which type of data independence is demonstrated? c) Why are the applications unaffected?	
OR		
(b)	A university currently maintains separate files for admission, fees, attendance, examination, and library management. The university is facing problems such as data redundancy, inconsistent records, difficulty in sharing data, security issues, and difficulty in generating integrated reports. Analyze the limitations of the file-processing system in this scenario and explain how a database management system can overcome each limitation.	
(10 marks)		CO1
Q2	Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period of time and has an associated due date, and the date when the payment was received	
(a)	a) Identify entities, attributes and relationships giving functionalities and draw E-R diagram for the system. b) Convert this to relational tables explaining logic involved	
OR		
(b)	i. Explain specialization and generalization with diagrams. ii. Suppose an EMPLOYEE entity is specialized into MANAGER, DEVELOPER, and TESTER. An employee can be both a developer and a tester. Which specialization constraint should be used? Explain your reasoning.	
(10 marks)		CO
Q3	A company stores employee information and their dependents. An employee has Employee_ID, while a dependent has Name, Age, and Relationship. A dependent cannot exist in the database unless the corresponding employee exists.	
(a)	Design the ER representation and identify the strong entity, weak entity, primary key, partial key, and identifying relationship and explain each of them.	
OR		
(b)	STUDENT(Student_ID, Name, Course, Age, City) COURSE(Course_ID, Course_Name, Credits, Department) ENROLLMENT(Student_ID, Course_ID, Semester, Grade)	

	1. Write a relational algebra expression to retrieve the names of all students enrolled in the BCA course. 2. Write a relational algebra expression to find the names and cities of students who belong to Dehradun. 3. Write a relational algebra expression to retrieve the details of students whose age is greater than 20. 4. Find the names of students who are enrolled in a course with more than 3 credits. Use appropriate relational algebra operations. 5. Write a relational algebra expression to find the Student_IDs of students enrolled in the course with Course_ID = 'C101'.	
Q4	(10 marks)	CO2
(a)	Explain the different steps briefly that are involved in processing a high-level query.	
	OR	
(b)	Q1 Consider the relational algebra expression: $\sigma_{Dept="CSE"}(Employee \bowtie Department)$ (a) Explain how the query is evaluated. (b) Apply query transformation rules to optimize it.	
Q5	(10 marks)	
(a)	Explain the concept of keys in relational databases. Differentiate between primary key, candidate key, foreign key, and super key with suitable examples. What are integrity constraints in a relational model? Describe domain, entity integrity, and referential integrity constraints and explain their importance in maintaining database consistency.	CO1,2
	OR	
(b)	Define Functional Dependency. Explain its role in relational database design. How do functional dependencies help in identifying redundancy in relations Consider the following relation: STUDENT(CourseID, StudentID, StudentName, CourseName, Instructor) Given functional dependencies: • CourseID $\rightarrow$ CourseName, Instructor • StudentID $\rightarrow$ StudentName i) Identify the candidate key(s) for the relation. ii) Is the relation in 2NF? Justify your answer. iii) Decompose the relation into 2NF if necessary.	